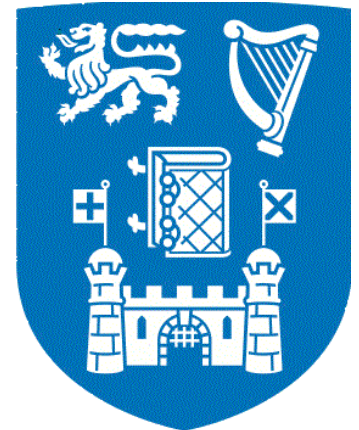


Formal Verification CSU44004/CSU55004

Lecture 2: Propositional Logic

Dr. Saif U. R. Malik
Assistant Professor, SCSS

School of Computer Science and Statistics
Trinity College Dublin



Recap

- ❖ Software: Error Prone, Failure can have significant impacts, Complexity of Software Designs
- ❖ Testing: Limitations
- ❖ Formal Methods: Myths, The Reality
- ❖ Formal Verification: Verification Methods, Algorithmic and Deductive Verification

Logic

- ❖ Logic is the science of correct reasoning
- ❖ If the train arrives late and there are no taxi's at the station, then John is late for his meeting.
 - ❖ John is not late for his meeting
 - ❖ The train did arrive late
- ❖ Therefore?
 - ❖ There were taxi's at the station

Propositional Logic

- ❖ Propositions are *declarative statements*.
- ❖ *Declarative statements* can be declared to be either **true** or **false** (but not both).
 - ❖ “Alice is an engineer”
 - ❖ “The sum of 3 and 5 is 8”
 - ❖ “The train is late”
- ❖ Not all statements are declarative. The following cannot be declared true/false.
 - ❖ “Let’s go to the cinema” (proposal)
 - ❖ “Where is Bob?” (question)
 - ❖ “Superb!” (exclamation)
 - ❖ “It will probably rain tomorrow” (likelihood)
- ❖ Propositional logic involves only declarative statements.

Propositional Logic

- ❖ Complex propositions can be constructed by simple ones using operators.
- ❖ p : “**If** *the train is late* **and** *there are no taxis in the station*, **then** *Bob is late to work*.”
- ❖ We can examine whether such propositions are true or false when we know the values of the basic propositions.
- ❖ English (or any human language) is imprecise and subtle (verb tenses, etc.) and error prone.
- ❖ Tom told Bill that he was wrong.
 - ❖ Who was wrong? Tom was wrong or Bill was wrong?
- ❖ A more mathematical language for logic would make the above arguments clear (stay tuned).
- ❖ $Told(Tom, Bill, Wrong(Bill))$

Propositional Logic

- ❖ Declarative statements have no intrinsic *truth-value*.
- ❖ They are simply a string of symbols, representing the declarative statements.
- ❖ All declarative statements (propositions) could be either True or False (but not both).
 - ❖ “Logic is interesting”
 - ❖ “This program terminates”
 - ❖ “Gates graduated Harvard”

Symbolic Propositional Logic

- ❖ We will consider declarative statements (propositions) to be *atomic* or *indecomposable*.
- ❖ We will name atomic propositions with *distinct mathematical symbols (lowercase English letters)*:
 - ❖ $p1, p2, \dots, q1, q2, \dots, r1, r2, \dots$
 - ❖ p =Logic is interesting
 - ❖ q =This program terminates
 - ❖ r =Gates graduated Harvard
- ❖ We form complex propositions using the following **operators**
 - ❖ **Negation**: symbols: $\neg \sim$ examples: $\neg p$ $\sim q$
 - ❖ **Disjunction**: symbols: $\vee$ **or** examples: $p \vee q$ $p' \text{ or } q'$
 - ❖ **Conjunction**: symbols: $\wedge$ **and** examples: $p \wedge q$ $p' \text{ and } q'$
 - ❖ **Implication or conditional**: symbols: $\rightarrow$ **implies**
 - ❖ examples: $p \rightarrow q$ $p' \text{ implies } q'$
 - ❖ **Parantheses**: symbols: $([])$

Symbolic Propositional Logic

❖ p: “The train is late”

❖ q: “There are taxis in the station”

❖ r : “Bob is late to work”

$$\text{❖ } (p \wedge \neg q) \rightarrow r$$

❖ p': “It is warm”

❖ q': “It is humid”

❖ r ' : “It is raining”

$$\text{❖ } q' \rightarrow p'$$

$$\text{❖ } (p' \wedge q') \rightarrow (r' \vee p')$$

$$\text{❖ } (p' \text{ and } q') \text{ implies } (r' \text{ or } p')$$

Symbolic Propositional Logic

❖ To not use so many parentheses we use **binding priorities**:

❖ $\neg$ binds more tightly than

❖ $\wedge$ and $\vee$ which bind more tightly than

❖ $\rightarrow$ which is right-associative

❖ eg:

❖ $(p1 \wedge (\neg q1)) \rightarrow [(r1 \vee (p2 \wedge (\neg q2))) \rightarrow r2]$

❖ Equivalent to

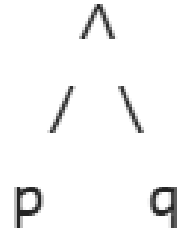
❖ $p1 \wedge \neg q1 \rightarrow r1 \vee (p2 \wedge \neg q2) \rightarrow r2$

Syntax

- ❖ Definition
 - ❖ The logical **Formulas** of Propositional Logic are exactly those accepted by the following grammar in Backus Naur Form (BNF):
 - ❖ $A, C, B ::= p \mid (\neg A) \mid (A \wedge A) \mid (A \vee A) \mid (A \rightarrow A)$
- ❖ Propositional Formulas can be represented as syntax trees where:
 - ❖ all the **leafs** are *atomic propositions*
 - ❖ and **all internal nodes** are *operators*
- ❖ Remember the binding priorities:
 - ❖ $\neg$ binds more tightly than
 - ❖ $\wedge$ and $\vee$ which bind more tightly than
 - ❖ $\rightarrow$ which is right-associative

Example Syntax/Parse Tree

❖ Formula (A) = $p \wedge q$



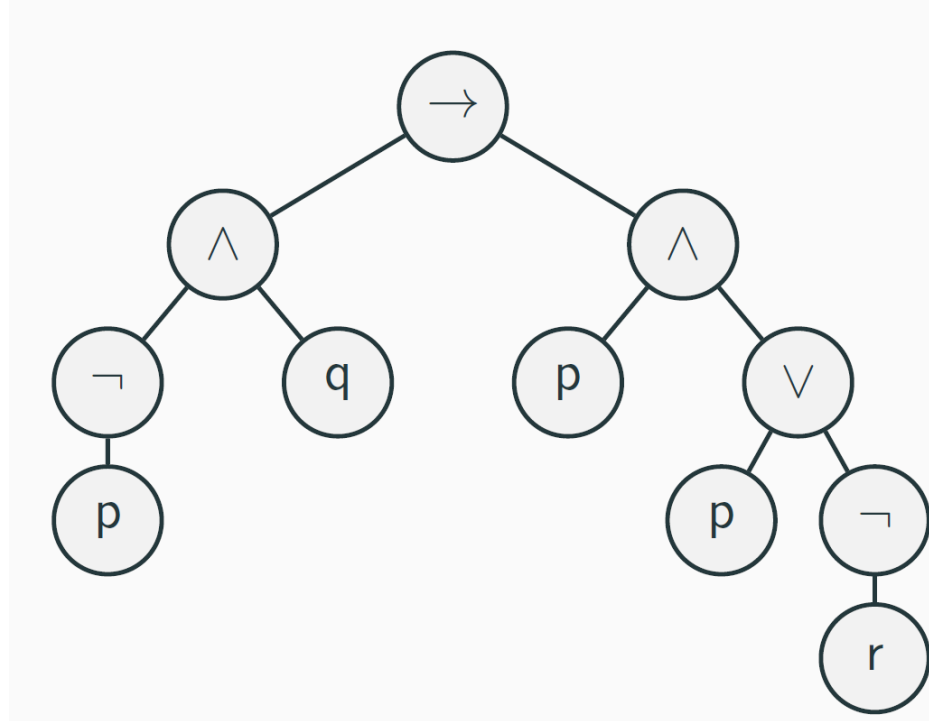
❖ Formula (A) = $\neg(p \vee q)$



Example Syntax/Parse Tree

❖ What is the parse tree of the formula:

❖ $(((\neg p) \wedge q) \rightarrow (p \wedge (q \vee (\neg r))))$



❖ With implicit binding priorities:

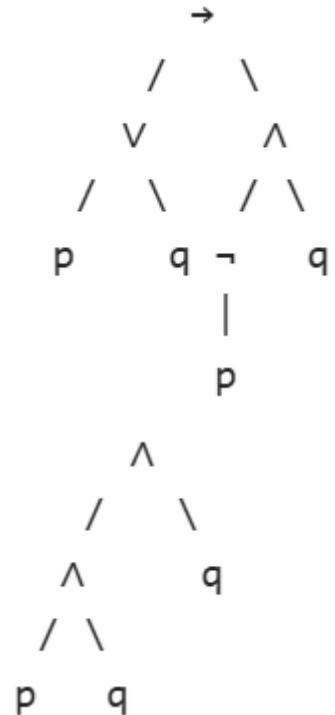
❖ $\neg p \wedge q \rightarrow (p \wedge (q \vee \neg r))$

Exercises

- ❖ Write in Symbolic Propositional Logic
 - ❖ 1. “Bob was jealous of Alice, or he was not in a good mood”
 - ❖ 2. “If the sun shines today then it won’t shine tomorrow”
- ❖ Can you parse the following formulas assuming the implicit binding?
 - ❖ 3. $p \vee q \rightarrow \neg p \wedge q$
 - ❖ 4. $p \wedge q \wedge q$

1) p: "Bob was jealous of Alice" q: "Bob was in a good mood" ($p \vee \neg q$)

2) p: "The sun shines today" q: "The sun shines tomorrow" ($p \rightarrow \neg q$)



Semantics of Propositional Logic Formulas

- ❖ Now we are going to look at the truth values of propositional logic **Formula A**. We will write this as $Sem(A)$
- ❖ Definition
 - ❖ **Truth values** contains two elements T and F, where T represents “true” and F represents “false”.
 - ❖ **Meaning of an atomic formula:** An atomic formula (propositional variable) such as p, q, or r is assigned exactly one truth value: $Sem(p) \in \{T, F\}$
 - ❖ Example: $Sem(p) = T$, or $Sem(p) = F$
 - ❖ **Valuation (Model):** A valuation (or model) is an assignment of a truth value to every proposition appearing in a formula.
 - ❖ Example: Formula $(A) = (p \wedge q) \rightarrow r$, its Model (A): $p=T, q=F, r=T$.
 - ❖ $p \wedge q = T \wedge F = F$
 $F \rightarrow T = T$
 - ❖ $Sem((p \wedge q) \rightarrow r) = T$

Semantics of Atomic Formulas

- ❖ The meaning of p : “It is warm” can be any of the following:
 - ❖ $\text{sem}(p) = T$
 - ❖ $\text{sem}(p) = F$
- ❖ A model of the formula $A = p$ can be any of the following
 - ❖ $p \mapsto T$
 - ❖ $P \mapsto F$

Truth Tables

- ❖ The meaning of operators $\wedge$ and $\vee$ are two functions with two arguments:

A	B	$A \wedge B$	A	B	$A \vee B$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	F	F	T	T
F	F	F	F	F	F

- ❖ A model of the formula $A = \neg p \wedge (q \vee p)$ can be any of the following

$$p \mapsto T, q \mapsto T$$

$$p \mapsto T, q \mapsto F$$

$$p \mapsto F, q \mapsto T$$

$$p \mapsto F, q \mapsto F$$

Truth Tables

- ❖ The meaning of operator $\rightarrow$ is a function with two arguments:

A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

- ❖ $A \rightarrow B \equiv \neg A \vee B$

Exercise

❖ Write the truth table of $\neg A \vee B \rightarrow B$

A	B	$\neg A$	$\neg A \vee B$	$\neg A \vee B \rightarrow B$
T	T			

A	B	$\neg A$	$\neg A \vee B$	$\neg A \vee B \rightarrow A$
T	T	F	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	T	F

Exercise

- ❖ Create the truth table of the formula:
 - ❖ $(p \rightarrow \neg q) \rightarrow (q \vee \neg p)$
- ❖ Give a model that makes the formula true.
- ❖ Give a model that makes the formula false.

Satisfiability - Validity

❖ Definition

- ❖ A formula (A) is *satisfiable* if there is at least one model that makes it true..
- ❖ A formula (A) is *falsifiable* if there is at least one model that makes it false..
- ❖ A formula (A) is *valid* (or a tautology) if it is true in every model..
- ❖ A formula (A) is a *contradiction/invalid* if it is false in every model..
- ❖ Q: which of the above are properties of
 - ❖ $(p \rightarrow \neg q) \rightarrow (q \vee \neg p)$
- ❖ Q: show that if $A1 \wedge (A2 \wedge A3)$ is satisfiable then $(A1 \wedge A2) \wedge A3$ is satisfiable.
- ❖ Q: Let $(A1 \wedge A2) \rightarrow A3$ be valid. Is it necessary that $A3$ is satisfiable?